ZKK CONSULTING - (REDACTED) FIVE-YEAR COMMERCIAL MODEL

Companion workbook to: Wrist-Worn Fall Detection - Market, Competitive, Patent and Commercial Analysis

Illustrative sample deliverable. Enterprise engagement + Patent Landscape Analysis. Data as of 23 August 2026.

HOW TO READ THIS WORKBOOK

Blue text on yellow fill = an input you can change. Every other cell is a formula and will update.

Black text = calculated in this sheet. Green text = pulled from another sheet in this workbook.

Column D on the Assumptions tab names the source or states that the figure is an unvalidated placeholder.

TABS

Assumptions Every driver in the model, in one place, each with its source or a flag that it has none.

Market Sizing TAM, SAM and SOM built from named public inputs rather than a single analyst headline.

BOM & Unit Cost Bill of materials from the client's own sourcing files, plus the cost lines those files omit.

Revenue Build Subscriber roll-forward and revenue by year, FY2027-FY2031.

P&L Gross profit, operating expense and EBITDA by year.

Scenarios Downside, base and upside on the four drivers the outcome is actually sensitive to.

Sources Every external figure used, with URL and retrieval date.

THE FOUR NUMBERS THAT DECIDE THIS BUSINESS

Customer acquisition cost, annual churn, the monitoring-centre cost per subscriber, and the contract-manufacture cost of the enclosure and assembly. None of the four could be sourced from public data. All four are marked as placeholders and all four are swept on the Scenarios tab. Get real quotes and real cohort data before this model is used for any decision that spends money.

LIMITATIONS

Projections are estimates based on the stated assumptions and are not forecasts, guarantees, or investment advice.

Actual results will differ. Figures are indicative and current as of the retrieval dates on the Sources tab.

This workbook does not constitute legal, regulatory, medical, tax or investment advice.

ASSUMPTIONS

Blue on yellow = editable input. Column D names the source, or flags the absence of one.

MARKET INPUTS

US adults aged 65+	61,200,000	persons	US Census Bureau, Vintage 2024 estimates (1 July 2024), released 26 Jun 2025
Share of adults 65+ living alone	26.0%	% of 65+	Pew Research Center, 4 Dec 2025, using Census/ACS via IPUMS (2023 data)
Wearable ownership, adults 50+	36.0%	% of 50+	AARP 2026 Technology Trends (n=3,838, fielded Sep-Oct 2025). Proxy for wearable receptivity; no 65+ figure is published
Current medical-alert penetration, 65+	9.0%	% of 65+	The Senior List 2026 Usage Report (n=909 adults 65+, fielded Apr-May 2026). Industry-sponsored survey; triangulates with Berg Insight
Category average monthly monitoring fee	$39.00	$/month	ConsumerAffairs medical alert statistics; SeniorLiving.org quotes a $25-$60 range (12 Aug 2026)

PRICING

Device price (one-time)	$199.00	$	Positioned at Bay Alarm SOS Smartwatch ($199) and Medical Guardian MGMove ($199.95), retrieved 23 Aug 2026
Subscription price, fall detection included	$34.95	$/month	Between Lively Basic ($24.99 + $9.99 FD) and Bay Alarm ($39.95 + $11 FD); well under UnaliWear Kanega ($64.95). Retrieved 23 Aug 2026
Device replacement cycle	3	years	ASSUMPTION - no published replacement-cycle data found for this category

HARDWARE COST PER UNIT

Bill of materials, qty 1,000	$13.28	$/unit	Client V3_PARTSCOMPS.xlsx, quantity-weighted, plus the 150mAh cell from All Parts and Components.xlsx. See BOM tab
PCB fabrication and assembly	$4.50	$/unit	PLACEHOLDER - no quote obtained. Requires a real assembler quote at target volume
Enclosure, strap and hardware	$6.00	$/unit	PLACEHOLDER - no quote obtained. Excludes tooling, which is captured separately
Packaging and literature	$1.50	$/unit	PLACEHOLDER - no quote obtained
Inbound freight and duty	$2.00	$/unit	PLACEHOLDER - no quote obtained. Tariff exposure on Chinese-assembled goods not modelled
Warranty and returns reserve	5.0%	% of unit cost	PLACEHOLDER - no category return-rate data found

SERVICE COST PER SUBSCRIBER

Cellular connectivity	$4.00	$/sub/month	PLACEHOLDER - benchmarked loosely against Verizon prepaid smartwatch retail ($10/mo); wholesale IoT rates not obtained

Monitoring centre (outsourced)	$6.00	$/sub/month	PLACEHOLDER - no public wholesale monitoring pricing found. Must be UL-listed for Medicaid channel access
Cloud, app and support	$0.75	$/sub/month	PLACEHOLDER - no quote obtained
Payment processing	3.0%	% of sub revenue	Standard card-present-absent merchant rate. Applied to subscription revenue

CUSTOMER ECONOMICS

Customer acquisition cost	$250.00	$/gross add	PLACEHOLDER - NO public CAC data exists for this category. The single most sensitive input in the model
Annual subscriber churn	22.0%	% per year	PLACEHOLDER - NO public churn data exists for this category. Second most sensitive input

GROSS SUBSCRIBER ADDITIONS BY YEAR

Fiscal year	FY2027	FY2028	FY2029	FY2030	FY2031
Gross adds	1,200	4,500	11,000	22,000	36,000

PLACEHOLDER ramp. Year 5 equals roughly 0.9% of the SAM on the Market Sizing tab. Not a forecast.

ONE-TIME COSTS (FY2027)

FCC / ISED / CE certification	$12,200.00	$	SparkFun published invoice for the Artemis module, 3 Dec 2019: FCC $4,000, ISED $3,000, Canadian agency $600, CE RED $2,800, CE LVD $1,800
Bluetooth SIG product qualification	$12,000.00	$	Bluetooth SIG fee schedule effective 1 Mar 2026, Adopter tier. Falls to $8,000 at Contributing Adopter ($3,500/yr membership)
Injection mould tooling	$45,000.00	$	PLACEHOLDER - no quote obtained

OPERATING EXPENSE (EXCLUDING SALES AND MARKETING)

Fiscal year	FY2027	FY2028	FY2029	FY2030	FY2031
Research and development	$180,000	$220,000	$260,000	$300,000	$340,000
General and administrative	$90,000	$140,000	$220,000	$320,000	$420,000

PLACEHOLDER structure. Replace with the client operating plan. Not derived from any external source.

MARKET SIZING - US, 2026

Built bottom-up from named public inputs. Estimates based on the stated assumptions; not forecasts.

TAM - THEORETICAL CEILING

US adults aged 65+	61,200,000	
Annual category spend per subscriber	$468.00	
TAM at 100% adoption	**$28,641,600,000**	**Theoretical ceiling only. No consumer category reaches 100% of a demographic.**

CROSS-CHECK - CURRENT US CATEGORY SPEND

Current subscribers, 65+ (bottom-up)	5,508,000	
Implied current US spend	**$2,577,744,000**	**Compare: Berg Insight puts North American telecare at US$3.7bn across 6.5m users, 2025. Two independent methods land in the same range, which is the strongest validation available here.**

SAM - SERVICEABLE ADDRESSABLE MARKET

Adults 65+ living alone	15,912,000	The population for whom an unwitnessed fall is the actual risk. Note this share has fallen from 29% in 1990 - the absolute number grows, the rate does not.
...that are wearable-receptive	5,728,320	Applies the 50+ wearable ownership rate as a receptivity proxy. Almost certainly generous: ownership among 65+ specifically is lower and is not measured in current public data.
Annual revenue per subscriber at our pricing	$485.73	Subscription plus the device amortised across its replacement cycle.
SAM	**$2,782,435,968**	

SOM - OBTAINABLE, FY2031

Ending subscribers, FY2031	55,565	
FY2031 revenue	$25,140,365	
SOM as share of SAM	**0.90%**	**A share this small is the point. In a category where the incumbent leader reports 900,000+ subscribers, a credible five-year plan is a rounding error on the market - and that is a reason for confidence in the plan, not against it.**

WHY THIS IS NOT A SINGLE ANALYST NUMBER

Published estimates of this category disagree by more than they agree. Three firms sizing "fall detection systems" in the same base year differ by 46% on level, and their 2035 forecasts differ by roughly six times ($857m against $5.17bn). The divergence is definitional: firms that count consumer smartwatches inside the category project a large market, and firms that exclude them project a small one. Averaging them would manufacture false precision. The build above uses government population data and disclosed-methodology surveys instead, and states every step.

BILL OF MATERIALS AND UNIT COST

Source: client files All Parts and Components.xlsx, LCSC_BOM.xlsx and V3_PARTSCOMPS.xlsx. Component prices as of the client's retrieval, March 2026.

V3 BILL OF MATERIALS - QUANTITY-WEIGHTED

Component	Qty	Unit @1	Unit @1k	Ext @1	Ext @1k	Note
MCU / BLE module	1	$3.6180	$2.2466	$3.6180	$2.2466	Row labelled ESP32-C3-MINI-1 but described as Nordic nRF52805 with an Espressif datasheet. Unresolved.
IMU / LSM6DSV16XTR	1	$5.6771	$4.1037	$5.6771	$4.1037	Upgraded from LSM6DSOXTR mid-programme. Largest single line.
Battery charger / MCP73831T	1	$0.7400	$0.3774	$0.7400	$0.3774	
USB-C receptacle, power only	1	$0.0493	$0.0200	$0.0493	$0.0200	
LDO regulator / XC6206P332MR	1	$0.1055	$0.0635	$0.1055	$0.0635	Absent from the earlier DigiKey list.
MOSFET / 2N7002	1	$0.0072	$0.0050	$0.0072	$0.0050	
TVS diode / ESD5Z5.0T1G	1	$0.0354	$0.0235	$0.0354	$0.0235	Absent from the earlier DigiKey list.
JST battery connector	1	$0.2119	$0.1285	$0.2119	$0.1285	Absent from the earlier DigiKey list.
Flyback diode / 1N4148WX	1	$0.0284	$0.0241	$0.0284	$0.0241	
Vibration motor / ERM coin	1	$0.5853	$0.3198	$0.5853	$0.3198	
Resistor 10k 0402	7	$0.0008	$0.0004	$0.0056	$0.0028	
Resistor 5.1k 0402	2	$0.0008	$0.0004	$0.0016	$0.0008	USB-C CC1 and CC2.
Resistor 100k 0402	1	$0.0008	$0.0004	$0.0008	$0.0004	
Capacitor 100nF 0402	4	$0.0052	$0.0027	$0.0208	$0.0108	
Capacitor 10uF 0603	1	$0.0065	$0.0038	$0.0065	$0.0038	
Capacitor 1uF 0402	1	$0.0041	$0.0024	$0.0041	$0.0024	
Capacitor 4.7uF 0402	1	$0.0034	$0.0015	$0.0034	$0.0015	
Subtotal - electronics				**$11.10**	**$7.33**	
Battery - 150mAh 3.7V Li-Po	1	$5.9500	$5.9500	$5.95	$5.95	Absent from both LCSC sheets. The qty-1,000 price in the client file equals the qty-1 price, which is almost certainly a copied cell rather than a volume quote. At $5.95 this line is 81% of the electronics cost beside it. Re-quote before use.
TOTAL BILL OF MATERIALS				**$17.05**	**$13.28**	

SOURCING MIGRATION - WHAT THE SAVING ACTUALLY WAS

Configuration	Qty 1	Qty 1,000	Note
V1 board, distributor sourcing (incl. battery)	$18.73	$14.06	Client All Parts and Components.xlsx. Excludes LDO, TVS and JST.

V1 board, re-sourced to assembler (excl. battery)	$5.39		Client LCSC_BOM.xlsx. 71.2% below the distributor figure at qty 1.
V3 board, assembler sourcing (excl. battery)	$11.10	$7.33	Quantity-weighted. Includes the upgraded IMU and the three added parts.
V3 board plus battery - like-for-like		$13.28	The only valid comparison against the $14.06 distributor figure.
Like-for-like saving at qty 1,000		**5.5%**	**The headline 71% is real, and it is a qty-1 figure on the original board. At volume, with the battery restored and the sensor upgraded, the saving is single digits. Both numbers are true; only the pair is honest.**

FULLY LOADED UNIT COST

Bill of materials (qty 1,000, incl. battery)	$13.28	
PCB fabrication and assembly	$4.50	
Enclosure, strap and hardware	$6.00	
Packaging and literature	$1.50	
Inbound freight and duty	$2.00	
Subtotal	$27.28	
Warranty and returns reserve	$1.36	
FULLY LOADED COST PER UNIT	**$28.65**	
Device price	$199.00	
Hardware gross margin	**85.6%**	**Four of the five cost lines above are unquoted placeholders. This margin is a structure, not a result.**

SUBSCRIBER AND REVENUE BUILD

Estimates based on the assumptions on the Assumptions tab. Not a forecast.

Fiscal year	FY2027	FY2028	FY2029	FY2030	FY2031	
SUBSCRIBERS						
Opening subscribers	–	1,068	4,838	13,564	30,160	
Gross additions	1,200	4,500	11,000	22,000	36,000	
Churned subscribers	(132)	(730)	(2,274)	(5,404)	(10,595)	
Closing subscribers	**1,068**	**4,838**	**13,564**	**30,160**	**55,565**	**Churn applied to opening balance plus half of gross adds (mid-year convention).**
Average subscribers	534	2,953	9,201	21,862	42,862	
REVENUE						
Device revenue	$238,800	$895,500	$2,189,000	$4,378,000	$7,164,000	
Subscription revenue	$223,960	$1,238,497	$3,858,839	$9,168,783	$17,976,365	
TOTAL REVENUE	**$462,760**	**$2,133,997**	**$6,047,839**	**$13,546,783**	**$25,140,365**	
Subscription share of revenue	48.4%	58.0%	63.8%	67.7%	71.5%	
UNIT ECONOMICS PER SUBSCRIBER						
Annual subscription revenue	$419.40					
Annual service cost	$141.58					
Annual service gross profit	**$277.82**					
Hardware gross profit (one-time)	$170.35					
Average subscriber life	4.5	years				
Lifetime value (undiscounted)	**$1,433.16**					
Customer acquisition cost	$250.00					
LTV to CAC ratio	**5.73x**					**Both inputs are placeholders. This ratio moves more than any other number in the workbook when they change.**
Months to recover CAC	3.4	months				

PROJECTED PROFIT AND LOSS - FY2027 TO FY2031

Estimates based on the stated assumptions. Not a forecast, a guarantee, or investment advice.

Fiscal year	FY2027	FY2028	FY2029	FY2030	FY2031	
REVENUE						
Device revenue	$238,800	$895,500	$2,189,000	$4,378,000	$7,164,000	
Subscription revenue	$223,960	$1,238,497	$3,858,839	$9,168,783	$17,976,365	
Total revenue	**$462,760**	**$2,133,997**	**$6,047,839**	**$13,546,783**	**$25,140,365**	
COST OF REVENUE						
Device cost	($34,379)	($128,920)	($315,137)	($630,274)	($1,031,358)	
Service cost	($75,605)	($418,094)	($1,302,676)	($3,095,219)	($6,068,502)	
GROSS PROFIT	**$352,776**	**$1,586,982**	**$4,430,026**	**$9,821,290**	**$18,040,505**	
Gross margin	76.2%	74.4%	73.2%	72.5%	71.8%	
OPERATING EXPENSE						
Sales and marketing	($300,000)	($1,125,000)	($2,750,000)	($5,500,000)	($9,000,000)	Gross adds multiplied by customer acquisition cost.
Research and development	($180,000)	($220,000)	($260,000)	($300,000)	($340,000)	
General and administrative	($90,000)	($140,000)	($220,000)	($320,000)	($420,000)	
One-time certification and tooling	($69,200)	–	–	–	–	FCC/ISED/CE, Bluetooth SIG qualification and mould tooling, all in year one.
Total operating expense	**($639,200)**	**($1,485,000)**	**($3,230,000)**	**($6,120,000)**	**($9,760,000)**	
EBITDA	**($286,424)**	**$101,982**	**$1,200,026**	**$3,701,290**	**$8,280,505**	
EBITDA margin	-61.9%	4.8%	19.8%	27.3%	32.9%	
Cumulative EBITDA	**($286,424)**	**($184,441)**	**$1,015,585**	**$4,716,875**	**$12,997,380**	
Peak cumulative cash requirement	**$286,424**					**Excludes working capital, inventory build and capital expenditure. A financing plan needs all three added.**

SCENARIOS AND SENSITIVITY

Change the four drivers on the Assumptions tab to the values below and read the result. Illustrative, not exhaustive.

THE FOUR DRIVERS THAT DECIDE THE OUTCOME

Driver	Downside	Base	Upside	Currently in model	Why it matters
Customer acquisition cost	$400.00	$250.00	$150.00	$250.00	No public CAC figure exists for this category. Sales and marketing is the largest operating line.
Annual churn	35.0%	22.0%	14.0%	22.0%	No public churn figure exists. Sets subscriber life, and therefore lifetime value.
Monitoring cost per subscriber	$9.00	$6.00	$4.00	$6.00	Unquoted. Sits directly in service gross margin, every month, forever.
Enclosure and assembly per unit	$18.00	$10.50	$7.00	$10.50	Unquoted. Small in absolute terms but it is most of the hardware cost that is not the bill of materials.

CURRENT MODEL OUTPUT

FY2031 revenue	$25,140,365
FY2031 EBITDA	$8,280,505
FY2031 EBITDA margin	32.9%
Peak cumulative cash requirement	$286,424
Lifetime value to CAC	5.73x

HOW TO USE THIS

This is a manual sensitivity table, not a data table, because the four drivers interact and a one-way table would understate the range. Set all four to the downside column, note the outputs, then set all four to upside and note them again. The gap between those two runs is the honest width of the uncertainty in this business case.

If the downside run does not survive, the problem is not the model. It is that the business plan depends on four numbers nobody has measured yet. Three of the four can be resolved with quotes inside a fortnight. Churn cannot - it takes a real cohort and real time.

RESULT OF RE-RUNNING THE MODEL AT EACH SETTING

Output (FY2031 unless noted)	Downside	Base	Upside	Comment
Total revenue	$22,349,869	$25,140,365	$27,080,662	Revenue barely moves. The subscriber plan is the same in all three.
EBITDA	($554,977)	$8,280,505	$14,437,815	Everything moves here. Same top line, opposite businesses.
EBITDA margin	-2.5%	32.9%	53.3%	

Closing subscribers	46,528	55,565	61,914	Churn is the only driver that touches the subscriber count.
Peak cumulative cash need	$4,244,176	$286,424	$141,954	A fifteen-fold swing. This is the number that decides whether the company needs to raise.
Lifetime value to CAC	2.13x	5.73x	15.53x	Below roughly 3x, paid acquisition stops working.
Months to recover CAC	11.8	3.4	0.0	

Computed by setting all four drivers to the named column and recalculating. Values are static; change the drivers on the Assumptions tab to reproduce.

THE ONE SENTENCE THAT MATTERS

The same subscriber plan produces either an $8.3m EBITDA business that funds itself on under $300k, or a business that loses money in year five and needs $4.2m along the way. Nothing separates those two outcomes except four numbers that have not been measured. Measuring them is cheaper than either outcome.

SOURCES

Every external figure used in this workbook. Retrieved 23 August 2026 unless otherwise stated.

Figure used	Source	Type	URL
61.2m US adults 65+ (1 Jul 2024)	US Census Bureau, Vintage 2024	Government	https://www.census.gov/newsroom/press-releases/2025/older-adults-outnumber-children.html
26% of 65+ live alone (2023)	Pew Research Center, 4 Dec 2025	Survey (Census/ACS)	https://www.pewresearch.org/short-reads/2025/12/04/a-smaller-share-of-older-us-adults-live-alone-today-than-in-1990/
36% wearable ownership, 50+	AARP 2026 Technology Trends	Survey (n=3,838)	https://www.aarp.org/pri/topics/technology/internet-media-devices/2026-technology-trends-older-adults/
9% medical-alert penetration, 65+	The Senior List 2026 Usage Report	Industry-sponsored survey	https://www.theseniorlist.com/research/medical-alert-device-consumer-usage-study/
$25-$60/month category range	SeniorLiving.org, 12 Aug 2026	Commercial publisher	https://www.seniorliving.org/medical-alert-systems/cost/
16.2m telecare users; NA US$3.7bn (2025)	Berg Insight via IoT Business News	Specialist analyst	https://iotbusinessnews.com/2026/02/28/the-number-of-telecare-solution-users-in-europe-and-north-america-reached-16-2-million-at-the-end-of-2025/
Bay Alarm SOS Smartwatch $199 + $39.95/mo	Bay Alarm Medical / SeniorLiving.org	Vendor / third party	https://www.bayalarmmedical.com/
Medical Guardian MGMove $199.95 + $42.95/mo	Medical Guardian via NCOA	Third party	https://www.ncoa.org/product-resources/medical-alert-systems/best-medical-alert-watches/
UnaliWear Kanega $299 + $64.95/mo	UnaliWear	Vendor primary	https://www.unaliwear.com/product/kanega-watch/
Lively $24.99/mo + $9.99 fall detection	Lively / Best Buy Health	Vendor primary	https://shop.lively.com/pages/compare-plan
Certification cost anchor $12,200	SparkFun published invoice, 3 Dec 2019	Primary invoice	https://news.sparkfun.com/3124
Bluetooth SIG qualification $12,000 (Adopter)	Bluetooth SIG fee schedule, eff. 1 Mar 2026	Standards body	https://www.bluetooth.com/develop-with-bluetooth/qualification-listing/qualification-listing-fees/
FCC grantee code fee $35	47 CFR 1.1103, eff. 23 May 2025	Federal Register	https://www.federalregister.gov/documents/2025/04/23/2025-06899/schedule-of-application-fees
Component prices and part numbers	Client files: All Parts and Components.xlsx, LCSC_BOM.xlsx, V3_PARTSCOMPS.xlsx	Client-supplied	Retrieved by client, March 2026. Quantity weighting applied by ZKK
Fall-detection market $469.5m-$684.6m (2025)	Future Market Insights / Grand View / Emergen	Analyst press summaries	See report Section 3.4 - used only to show the disagreement, not as an input
Connect America 900,000+ subscribers	Connect America corporate site	Vendor primary	https://www.connectamerica.com/about/

NOT SOURCED - PLACEHOLDERS IN THIS WORKBOOK

Customer acquisition cost · Annual churn · Monitoring-centre cost per subscriber · Cellular wholesale rate ·

Cloud and support cost · PCB assembly · Enclosure and strap · Packaging · Freight and duty · Warranty reserve ·

Injection mould tooling · Device replacement cycle · Research and development and general and administrative expense ·

The gross-add ramp · Component price curve with volume (BOM held flat at the qty-1,000 quote at all volumes) ·

Device replacement revenue (the 3-year cycle drives the SAM only, not the revenue build).

None of these could be established from public data. All editable inputs are marked in blue on yellow.